Image Generation Models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach diffusion-style intuition within the stated toy scope.

Supply both native evidence cards, keep cases separate and assess the fresh diagram record before feedback.

Keep one habit: Require visible requirements and provenance; do not infer denoising quality or a real event from appearance.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of image generation and **provenance**

Two 45-minute sessions. An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can describe a toy **denoising process** and distinguish generated illustrations from **observation**s.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-3/10, ai-3/11. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

Look for: Sample from a distribution using explicit cumulative intervals

Reference example, not the only acceptable wording: Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open [0,1) **input** contract and must be rejected.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Prompt Evaluation

Iterative Prompting

Sampling and Variability

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

Concept to explain

Image generation iteratively constructs pixels guided by learned representations.

Term	Meaning
latent representation	An internal representation learned or constructed by a model rather than directly supplied as the original observation.
noise	Variation or interference that obscures the information relevant to a task.
denoising	Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs.
conditioning	Making a model's output depend on supplied information, such as a prompt or image.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach one image-generation family without implying all generators work alike

> Keep the grid-process illustration and poster-evidence audit distinct.

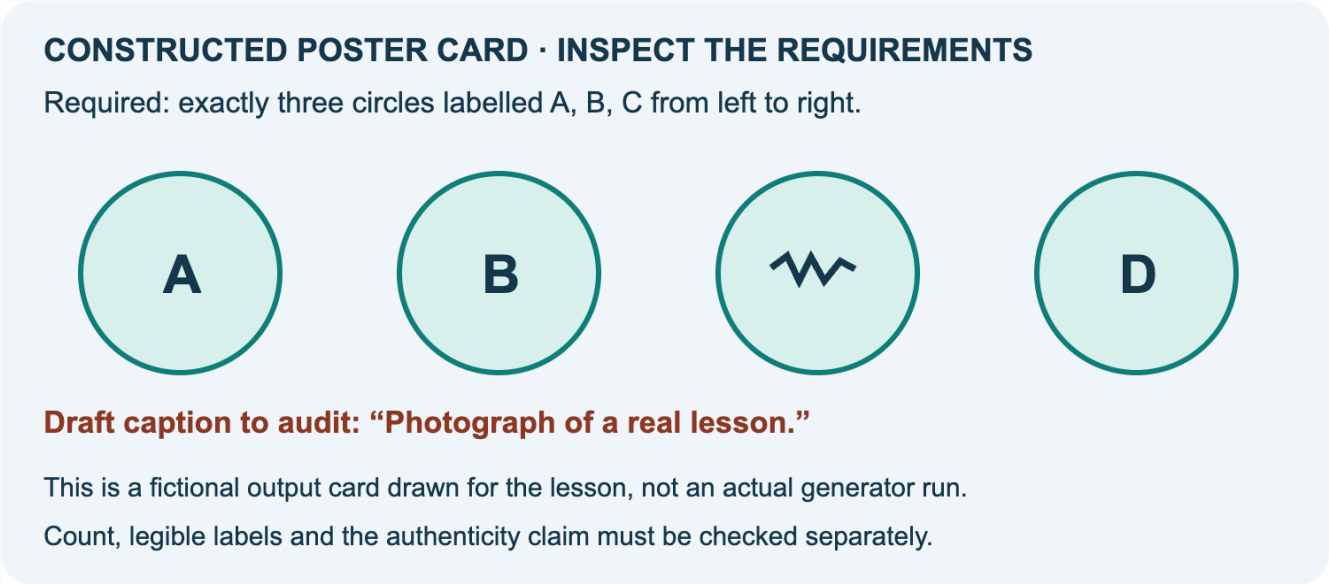
State generation and provenance boundaries

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Inspect the complete constructed poster



A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Read the supplied toy grids

A fictional lesson poster illustrates one family of image-generation approaches.

Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

TWO SUPPLIED TOY GRIDS · ONE ILLUSTRATIVE UPDATE



These arrays are invented teaching values, not trained-model outputs.

No target image or learned update rule is supplied; do not infer measured denoising quality.

Trace the imagined update and architecture scope

AI ACADEMY / AI-3/10 · 02 · TRACE THE CASE

Follow the deciding steps

01 Noisy representation

Four toy numerical cells form the starting grid.

02 Condition on request

A text request guides the imagined learned denoising process.

03 Illustrative update

Use the supplied second grid to show one change in representation.

04 Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Noisy representation	Four toy numerical cells form the starting grid.
2. Condition on request	A text request guides the imagined learned denoising process.
3. Illustrative update	Use the supplied second grid to show one change in representation.
4. Check provenance	Caption it as a generated-process illustration, not a photographed experiment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Reject the photographed-measurement caption

Change: Relabel the illustration as a photograph of an actual laboratory measurement.

Prediction: That caption is unsupported.

Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

State what the illustration does not establish

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Technical reference: Rombach et al., High-Resolution Image Synthesis with Latent Diffusion Models, describes diffusion in a learned latent representation with conditioning such as text. The classroom arrays do not implement or evaluate that model. Primary source: https://arxiv.org/abs/2112.10752

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. In this simplified diffusion-style image generator, begin with random visual **noise**. Other image-generation methods use different procedures.

2. In this simplified account, use a learned estimate to update the noisy representation, with the request providing conditioning. Individual steps need not visibly improve.

3. Learned patterns influence shapes, textures, and relationships.

4. The result is generated, not retrieved **evidence** of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

Exact explanation to model

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Provenance	Recorded origin and transformation history
Visual criterion	A stated requirement checked against an image
Authenticity claim	A claim about the source or reality of depicted content

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Image Generation Models

The facts we will use

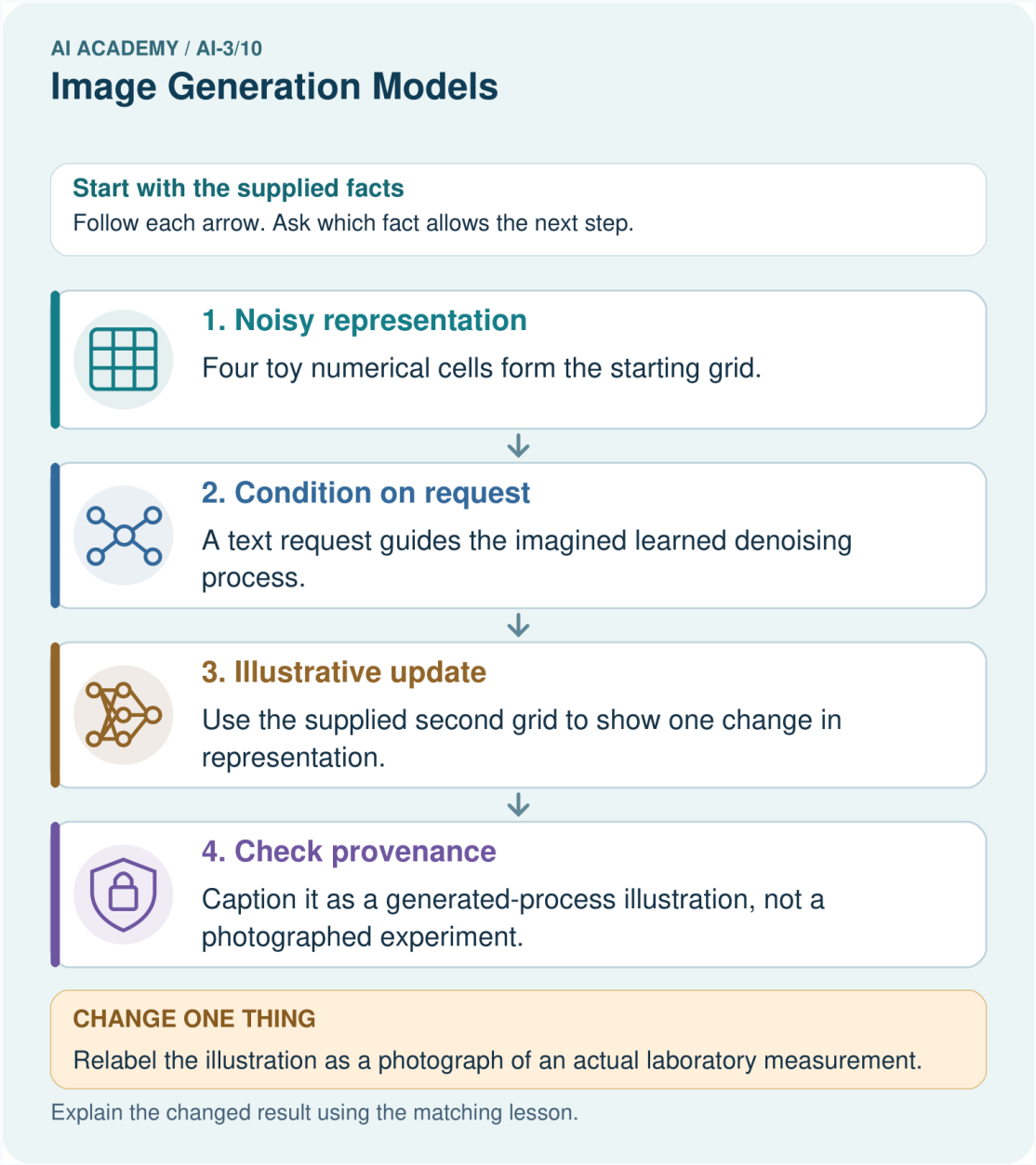
AI ACADEMY / AI-3/10

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What may these two grids explain, and what can they not establish about a real generator?

They explain an illustrative staged change. They cannot establish learned weights, actual image quality or accurate scientific **observation**.

Changed-case reasoning: That caption is unsupported. Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A generator creates a realistic fictional bicycle beside a tree.

Write the question to explain: How does this differ from event photography?

Reveal after **prediction**s: It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

Use the earlier changed question as optional later practice. Reserve Audit a new diagram independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Process cards: request K and initial representation L feed imagined update M; N records the absence of an executed model, final poster or quality score. Grid differences -0.1,+0.1,-0.2,-0.1 do not establish reduced error. The separate poster has four rather than three circles; its third label is illegible and D is not required C. The photograph caption is unsupported.

Label it a constructed classroom card, not an actual generator output. Independent two-arrow record: count passes, swapped labels fail, and generation metadata does not support an experimental-observation caption.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Use the supplied process cards, numerical grids and separate constructed poster card. Trace inputs and illustrative update, then check visible counts/labels and provenance. No real generator is run and no final image or denoising-quality measurement follows from the arrays.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit a new diagram independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A generated diagram requires two arrows and labels **input/output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

W2 Trace the worked case

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

W3 Audit the limit

A report omits this qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

Additional transfer practice question

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

Reasoned practice answer

Generated images can miss exact counting constraints. Use a checked, editable drawing or diagram tool for an instructional diagram requiring exactly four roots.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “An image generator always obeys numerical constraints precisely.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

It demonstrates a generated depiction, not that a greenhouse exists on Mars. Label it as generated and separate design imagination from factual **evidence**.

Some image models use iterative **denoising**, but no single architecture describes every generator.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A generated diagram requires two arrows and labels **input/output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

Teacher explanation: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Treat generated diagrams as illustrations; pair factual claims with approved sources.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Arrow count passes; label semantics fail. Metadata supports generation **provenance**, not observed-event authenticity; replace the caption and correct labels. Require intermediate reasoning and the stated scope.

Fresh retry: A generated map has three required labels but omits one and is captioned survey **evidence**. Expected reasoning: Completeness fails; generation cannot establish surveyed geography.

Use verified map data for geographical claims. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed check

Suggested return points in this level: ai-3/11, ai-3/13, ai-3/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 11: Visual Composition. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week11: Visual Composition. Carry forward visible count/label criteria, source-grounded meaning and an honest production record when planning layout and checking a diagram.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of image generation and **provenance**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Check every learner before discussion: Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Relabel the illustration as a photograph of an actual laboratory measurement. Require a **prediction** and a reason before comparing. Expected result: That caption is unsupported.

Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

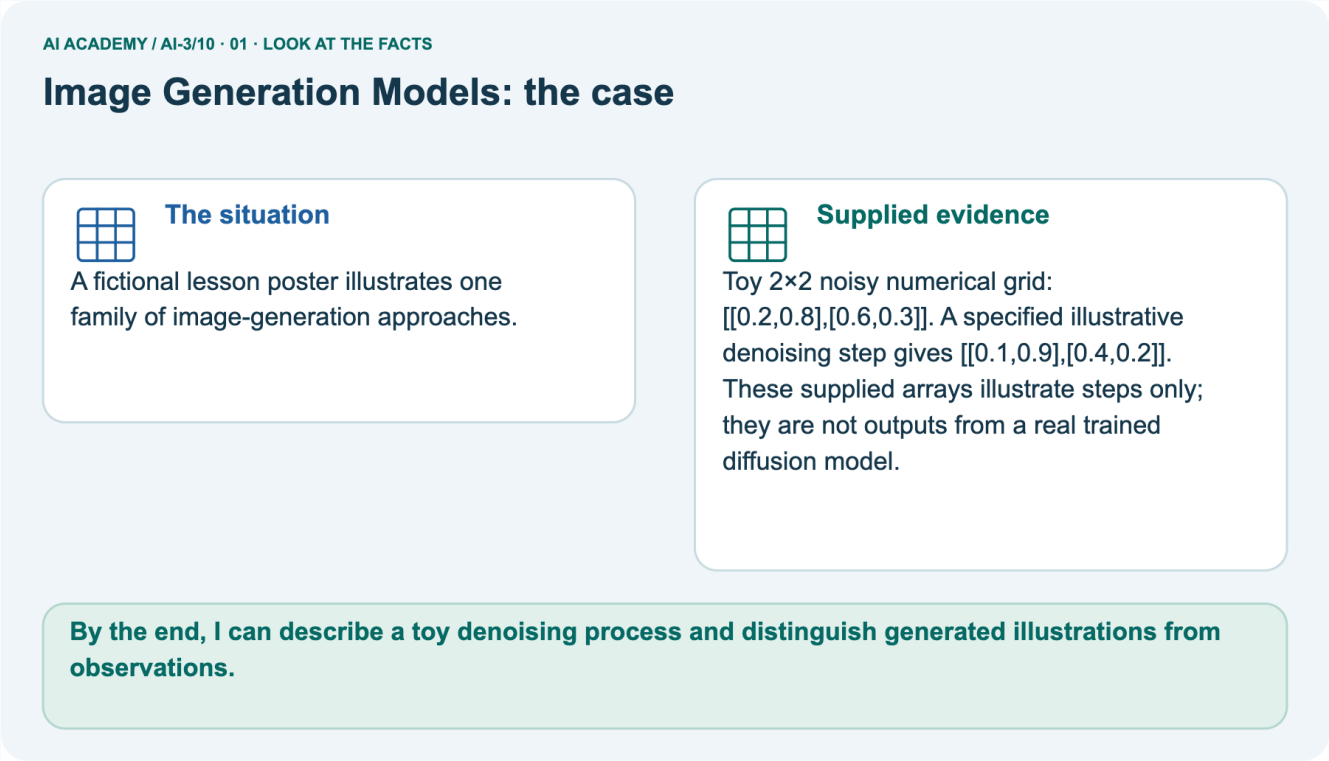
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Image Generation Models: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

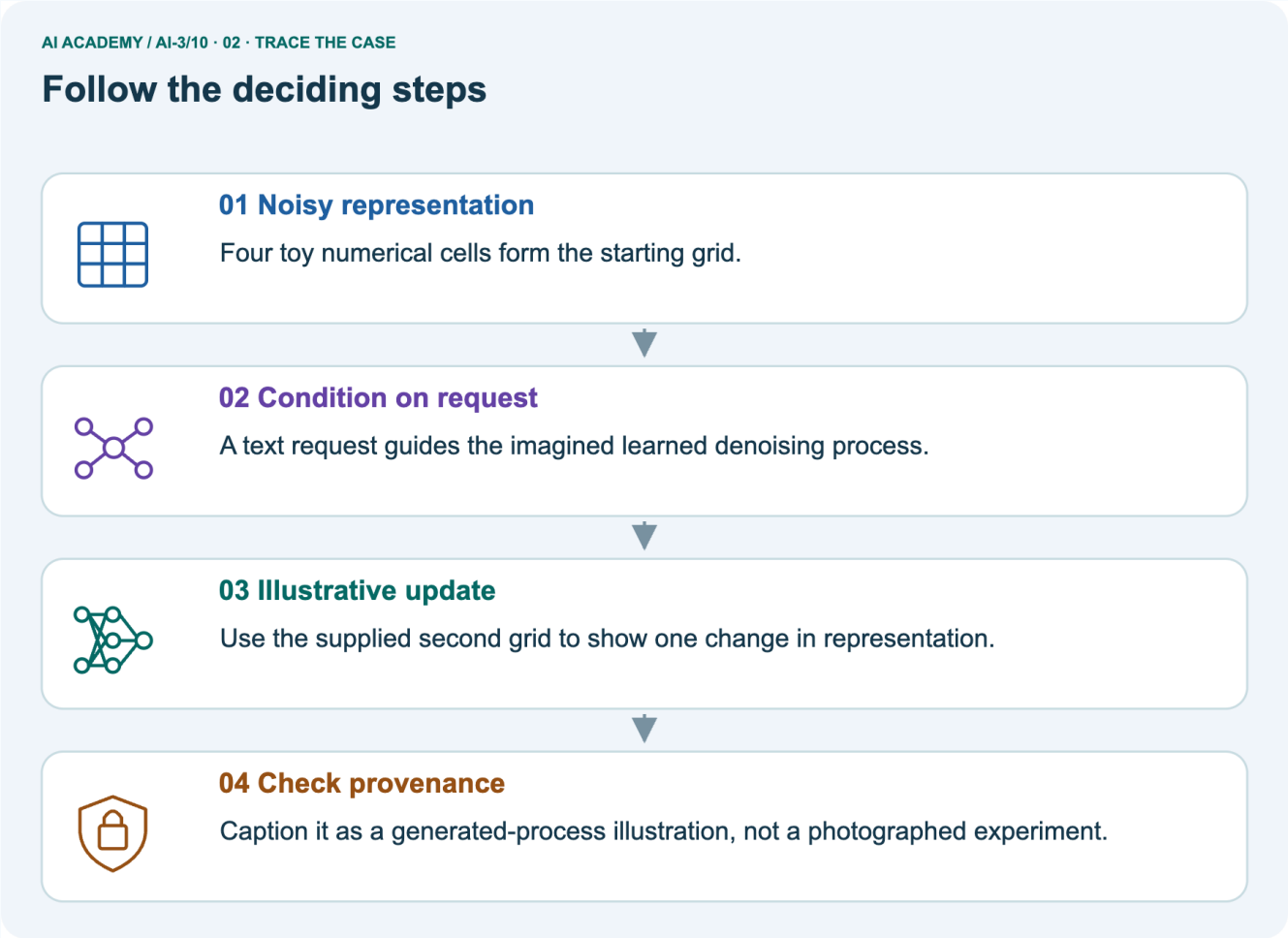
A fictional lesson poster illustrates one family of image-generation approaches. Supplied facts: Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



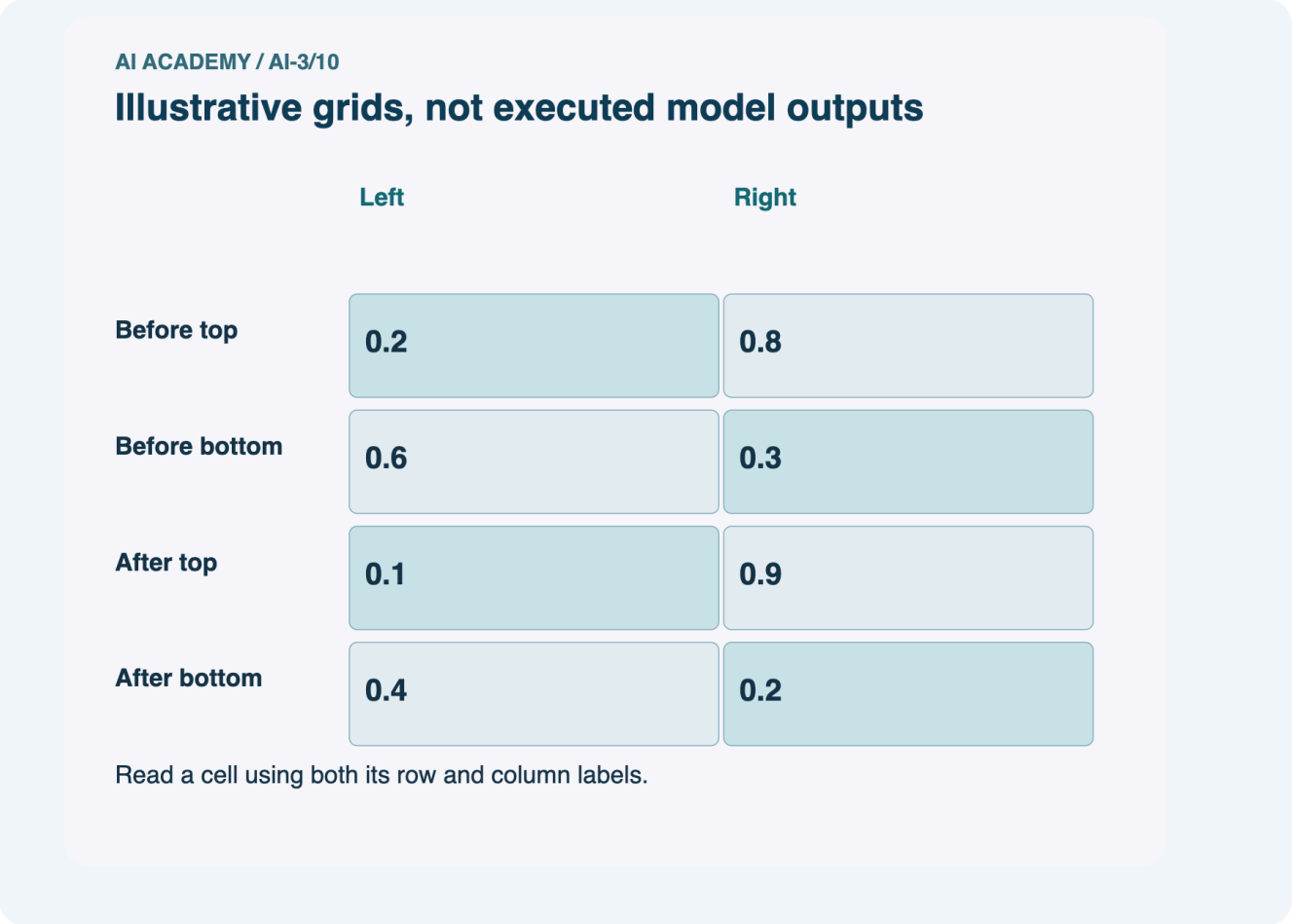
Follow the labelled picture with the matching explanation. Open the vector diagram

Noisy representation: Four toy numerical cells form the starting grid. → Condition on request: A text request guides the imagined learned **denoising process**. → Illustrative update: Use the supplied second grid to show one change in representation. → Check **provenance**: Caption it as a generated-**process** illustration, not a photographed experiment.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Illustrative grids, not executed model outputs



Follow the labelled picture with the matching explanation. Open the vector diagram

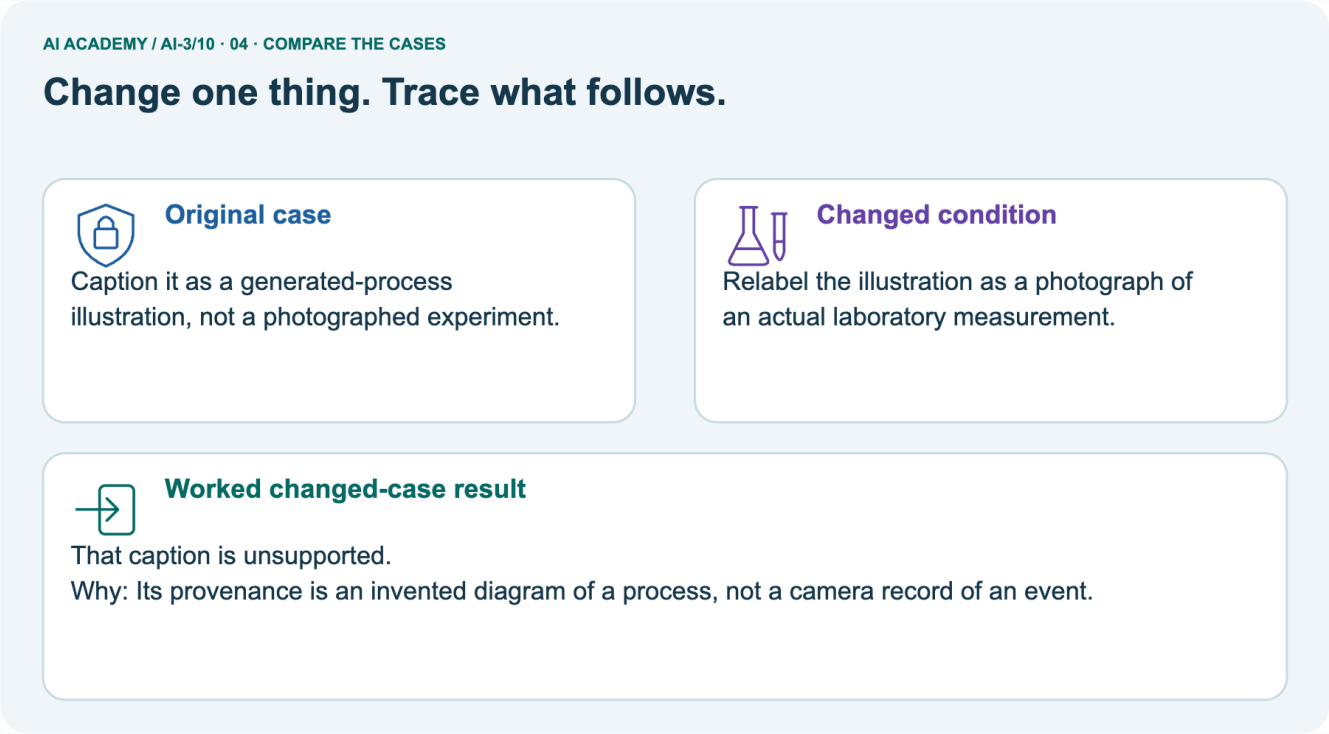
Illustrative grids, not executed model **output**s. Before top: Left = 0.2, Right = 0.8. Before bottom: Left = 0.6, Right = 0.3. After top: Left = 0.1, Right = 0.9. After bottom: Left = 0.4, Right = 0.2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Caption it as a generated-**process** illustration, not a photographed experiment. Changed condition: Relabel the illustration as a photograph of an actual laboratory measurement.

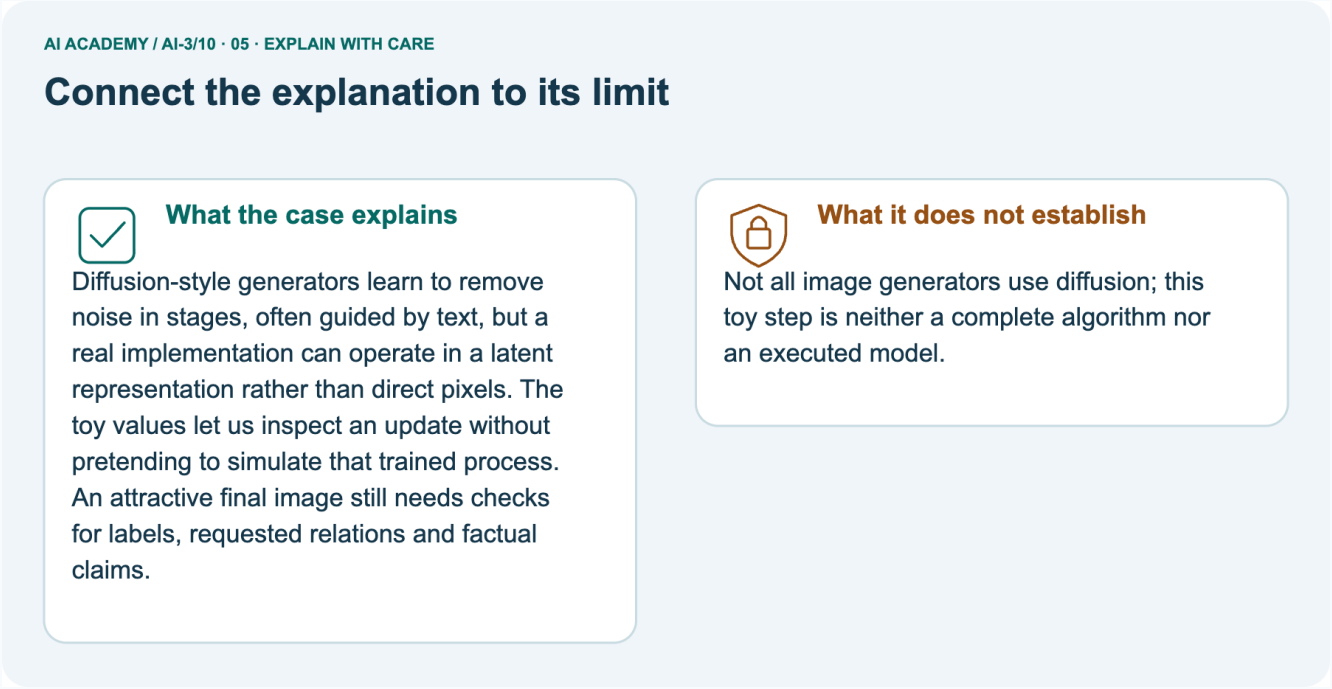
Expected result: That caption is unsupported. Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

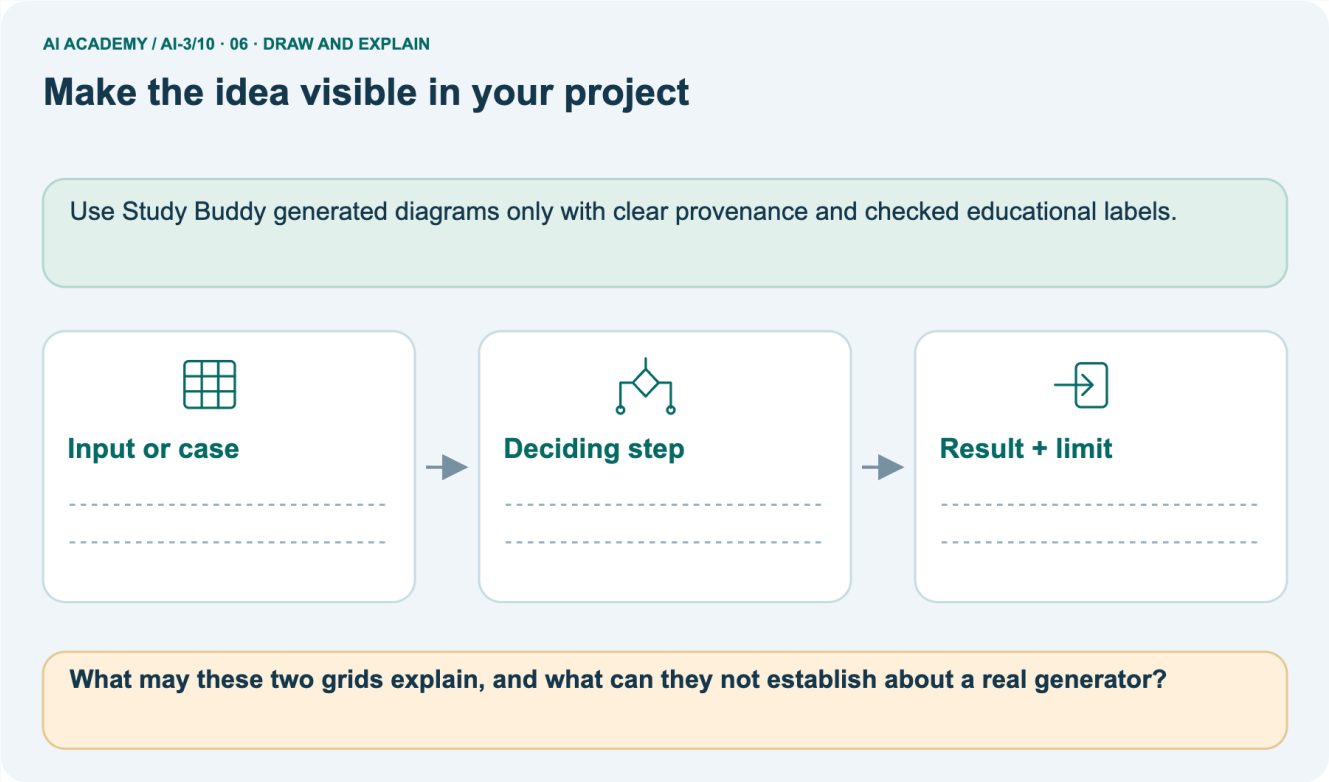
Limit: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels. Draw your **input**, deciding step, result and limit.

Supported question: What may these two grids explain, and what can they not establish about a real generator?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: They explain an illustrative staged change. They cannot establish learned weights, actual image quality or accurate scientific **observation**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Image Generation Models: the case

02. Follow the deciding steps

03. Illustrative grids, not executed model outputs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.